Neural Weight Generation for Portfolio Replication

Fintech Group Project 3

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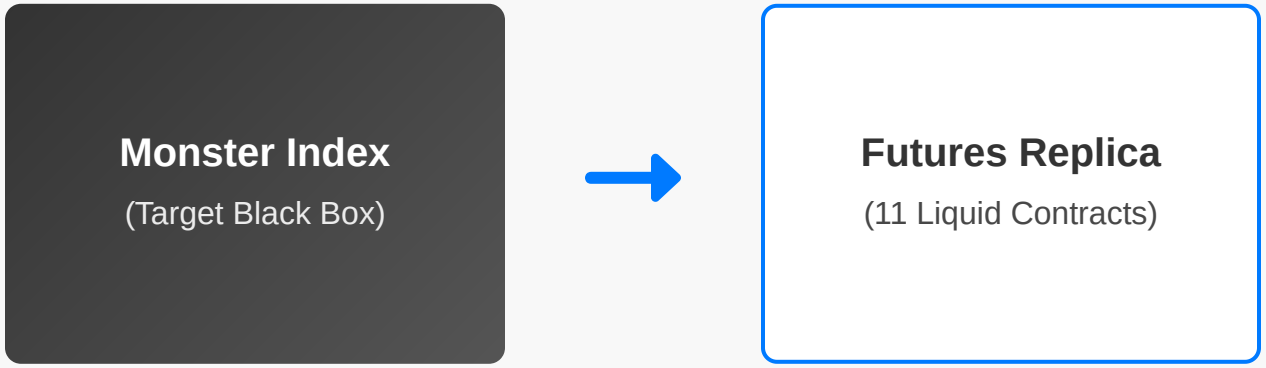
Replicating Complex Portfolios with Liquid Futures

Our objective is to replicate a complex "Monster Index" (50% HFRX Hedge Fund Index, 25% MSCI World, 25% Global Agg Bond) using only 11 highly liquid Futures contracts.



Neural Networks Enable Dynamic Weight Adaptation

Dynamic Weight Generation

Unlike static linear models (OLS, Ridge, LASSO), neural weight-generators learn **dynamic weights** that adapt to evolving market regimes, offering greater flexibility.

Focus on Weights, Not Returns

The neural network outputs portfolio weights (β_t), explicitly NOT return predictions. This design choice keeps portfolio arithmetic external, enhancing interpretability and control.

$$replica_return = X_t \cdot \beta_t$$

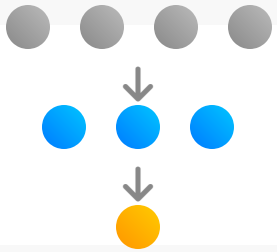
$$\beta_t = NN(recent\ futures\ window)$$

Where X_t represents the returns of the 11 liquid futures contracts at time t.

Comparative Architectures: MLP vs. LSTM

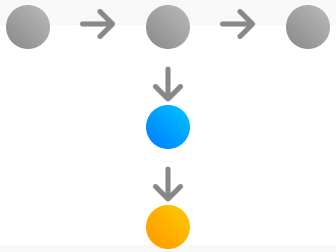
Multi-Layer Perceptron (MLP)

Flattens the input window, processing data through fully connected dense layers. Efficient for static feature sets.



Long Short-Term Memory (LSTM)

Processes sequences step-by-step, inherently capturing temporal dependencies and long-range patterns.



Unconstrained Outputs: Allows for both long and short positions in the portfolio.

UCITS Compliance: VaR scaling applied post-inference to meet regulatory limits.

Risk Limit: Ensures 1M VaR 99% ≤ 20% for robust risk management.

VaR Method: Historical Simulation — current weights projected onto last 52 weeks of factor returns; empirical 1st percentile read directly. No normality assumption. More conservative than parametric VaR.

Robust Training Setup and Model Configurations

LOSS & OPTIMIZATION

Optimized using **Adam** with gradient clipping and early stopping to ensure convergence and prevent overfitting.

DATA STRATEGY

60% Train / 15% Val / 25% Test

Chronological split maintained to prevent look-ahead bias and data leakage, reflecting real-world deployment conditions.

MODEL CONFIGURATIONS

Config	Input Window	Hidden Layers	L1 Penalty (λ)
MLP-S	26 Weeks	[64, 32]	0
MLP-L	52 Weeks	[64, 32]	0.001
MLP-D	26 Weeks	[128, 64, 32]	0.001
LSTM	52 Weeks	[64]	0

Note: fixed 60/15/25 train/val/test split. Full walk-forward retraining at each step is computationally prohibitive for NNs; acknowledged as a departure from the rolling-window standard used by linear models.

Out-of-Sample Performance (Test Set)

Config	Tracking Error	Info Ratio	Correlation	Sharpe	Max Drawdown
MLP-S	3.63%	-0.238	0.842	0.841	10.0%
MLP-L	5.12%	-0.527	0.687	0.919	4.8%
MLP-D	5.02%	-0.509	0.683	0.837	5.7%
LSTM ★ ✅ BEST	**3.44%**	**-0.227**	**0.862**	**0.890**	**9.6%**

Best config selected by Information Ratio. Negative IR is expected — the monster index contains hedge fund alpha unobservable from Futures returns alone.

⚠️ *MLP-L anomaly: Sharpe 0.919 despite worst tracking — the L1 penalty collapses weights to near-cash (gross exposure 10–25%), reducing variance without tracking the target. High Sharpe ≠ good replication.*

Why LSTM Wins: Sequential Processing and Generalization

Sequential Processing

LSTM's 52-week recurrent processing captures **regime dynamics** (momentum, risk-off shifts) that a flattened MLP encodes only as static positional features. MLP-S (best MLP) has the same architecture depth but a shorter 26-week window — the gap to LSTM is attributable to sequential memory.

Better Risk Profile

The LSTM achieves **LOWER maximum drawdown (9.6%)** than MLP-S (10.0%) while also delivering better tracking error — stable, diversified weights provide both replication quality and genuine risk efficiency.

Strong Directional Tracking

Rolling 26-week correlation stable at **0.85–0.97** throughout; brief dip to ~0.62 during March 2020 (linear factor structure breakdown in tail events) before full recovery.

0.862

CORRELATION

Strong directional tracking of the target portfolio.

3.44%

TRACKING ERROR

Lowest out-of-sample error across all tested configurations.

LSTM Learns Regime-Aware Allocations

Stable, Interpretable Portfolio Structure

The LSTM learns a remarkably stable allocation:

- Long ES1 (S&P 500): **~+16%**
- Long TY1 (US Treasury 10Y): **~+13%**
- Short DU1 (German Schatz 2Y): **~-28% average**, deepening to -38% in March 2020

The DU1 structural short reflects historically negative German short-term yields relative to US equivalents. The deepening of this short during March 2020 — followed by normalisation — shows the LSTM recognising the flight-to-quality environment from within-window return dynamics alone.

Gross exposure stable at **~90%**, well below the UCITS 200% limit.

Overfitting Analysis Across Configurations

Three Distinct Failure Modes

⚠ MLP-D (Capacity Overfitting)

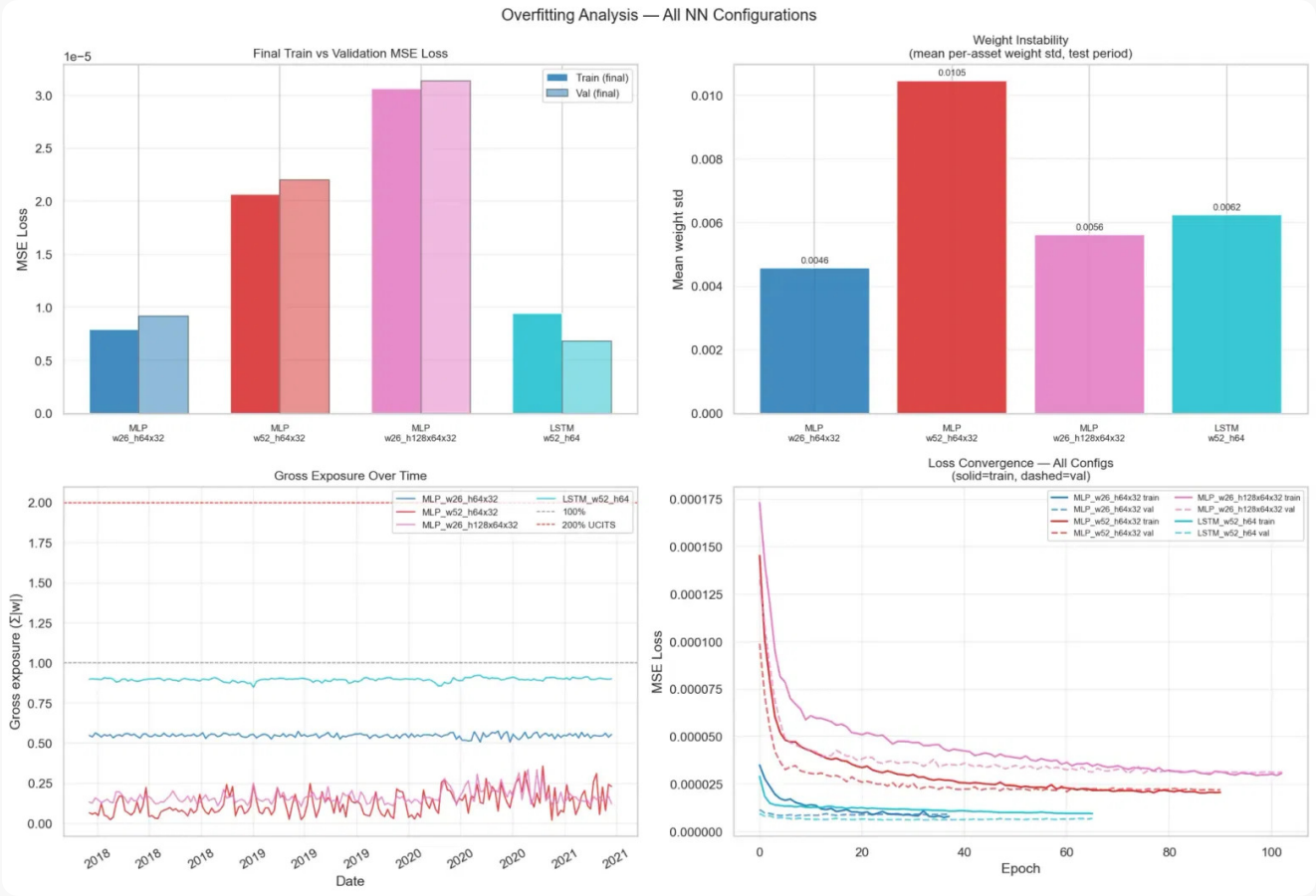
Train MSE << Val MSE — classic overfitting signature. Weight instability 0.0105, 2.3× higher than MLP-S. The L1 penalty reduces gross exposure but does NOT stabilise temporal weight dynamics.

✕ MLP-L (Regularisation Collapse)

L1 penalty on 572 flattened features drives near-cash solution (gross exposure 10–25%). Eliminates overfitting by eliminating the portfolio. High Sharpe is an artefact of near-zero variance.

✓ LSTM (Regime Shift, NOT Overfitting)

Val loss < Train loss throughout — the ONLY config with this pattern. Training period (2007–2015) includes GFC, European debt crisis. Validation period (2015–2018) is a low-volatility bull market.



Top-left: final train/val MSE. Top-right: weight instability score.
Bottom-left: gross exposure by config. Bottom-right: all convergence curves.

Current Limitations and Future Research

CURRENT LIMITATIONS

- **Data Constraints:** Small dataset size (~420 training samples) limits the ability to train deeper architectures effectively.
- **Feature Scope:** Current model relies solely on price action; absence of macro-economic features or sentiment indicators.
- **Walk-Forward Gap:** NN uses a fixed 60/15/25 split while linear models retrain at every step — comparison is partially unfair. Full walk-forward NN retraining at weekly frequency is computationally prohibitive.

NEXT STEPS

- **Volatility Signals:** Incorporate realized and implied volatility as additional input features to improve regime detection.
- **Turnover Penalty:** Add a regularization term to the loss function to penalize excessive rebalancing and reduce costs.
- **Benchmarking:** Compare neural weight generation against classical state-space models like the Kalman Filter.
- **Transaction Costs:** Analysis confirms cost drag is negligible (0.07% p.a. at 5 bps) due to low weekly turnover (~0.028 mean). UCITS VaR constraint never binding — peak historical-simulation VaR ~8% vs 20% limit.

Thank You

Neural Weight Generation for Portfolio Replication

github.com/giuliatalà/portfolio-replication